WORKFLOW GUIDE

KVCET Hostel Management System

How each portal works and how records move between Students, Advisers, HODs, Wardens, Security, AO, Mess and the Dean.

Student | Adviser | HOD | Warden | Security | AO | Mess | Dean | Admin

Version as of 18 October 2026 · Screenshots taken from the running application with clearly-labelled demo data
Karpaga Vinayaga College of Engineering & Technology

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1 · OVERVIEW

One shared record, many portals

Every portal reads and writes the same database. A leave request the student submits is the same record the adviser recommends, the warden approves, security scans and the Dean sees in attendance monitoring. Nothing is copied between portals.

Student
Leave, outing, outpass QR, complaints, attendance, mess menu

Adviser
Recommends advisee leave, academics, mentoring & follow-ups

HOD
Department students, advisers, academics, follow-ups, incidents

Shared database · users · students · requests · outpasses · gate movements · attendance · complaints · incidents · tasks · schedules · reports · notifications · audit log

Warden
Approvals, attendance & absences, study sessions, rooms, incidents

Security
Scans outpass QR at the gate; records exit and return

AO
Incidents, escalations, maintenance work orders

Mess
Menu drafting & publishing, student feedback

Dean
Monitoring, assignments, schedules, report distribution, analytics

Admin
Accounts, students, departments, adviser assignment, audit log

Every record carries

- **A readable ID**, for example `LR-2026-0001` for a leave request or `FU-2026-0003` for a follow-up.
- **A status history**: who changed it, when, and with what remarks. It is visible in each record's timeline.
- **Notifications** to the next person who must act, linking straight to the record.
- **An audit-log entry** for every important action, visible to the Admin.

2 · SIGN-IN

Sign-in and access control

Each person has one account with one role. The role is stored on the server, which decides which portal and which records the person can open.

ADMIN
Create account
Adds staff or students with a temporary password

›

ANY USER
Sign in
Chooses their role, enters email and password

›

SYSTEM
Verify
Checks the password and that the role matches the account

›

FIRST LOGIN
Change password
Must set their own password before continuing

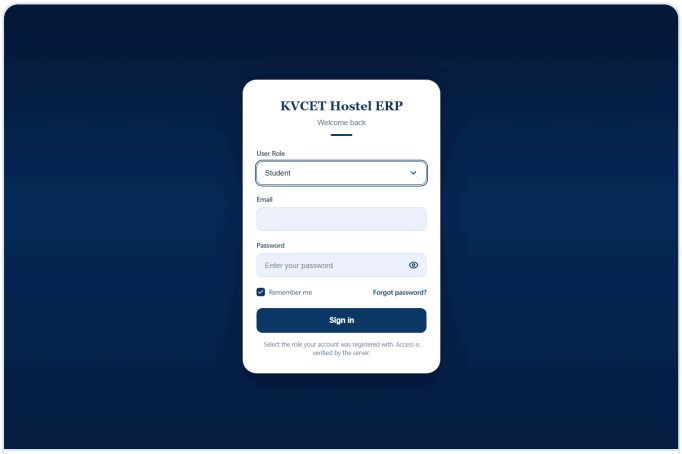
›

PORTAL
Own portal opens
Other portals are refused

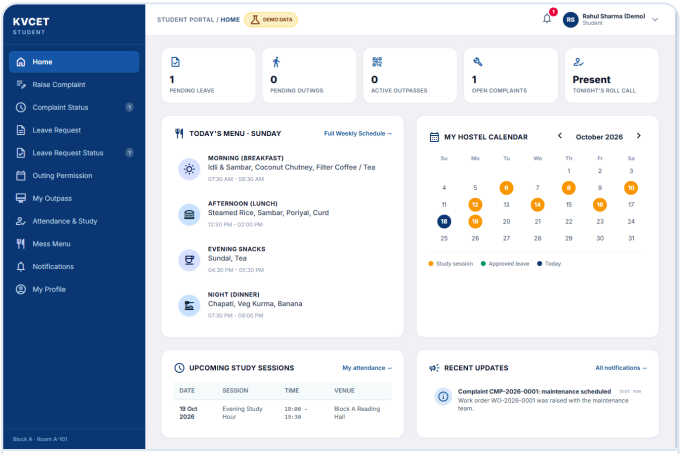
What each role can see

Role	Student records visible
STUDENT	Only their own requests, outpasses, attendance and complaints
ADVISER	Only their assigned advisees
HOD	Students of their own department
WARDEN AO DEAN ADMIN	All hostel students (as needed by their role)

- After 5 wrong passwords, the account is locked for 15 minutes.
- Sessions end after 2 hours of inactivity, or on sign-out.
- Typing another portal's address in the browser shows an "unauthorized" screen, and the server refuses the data.



Sign-in page. Access is verified by the server.



Student portal home after sign-in

3 · LEAVE

Leave request → digital outpass → gate

A leave goes through two approvals. The outpass is created only after the warden approves, and Security records the student's exit and return against that same pass.

STUDENT
1. Apply
Leave type, dates, reason, optional document

ADVISER
2. Recommend
or reject with a reason

WARDEN
3. Approve
or reject with a reason

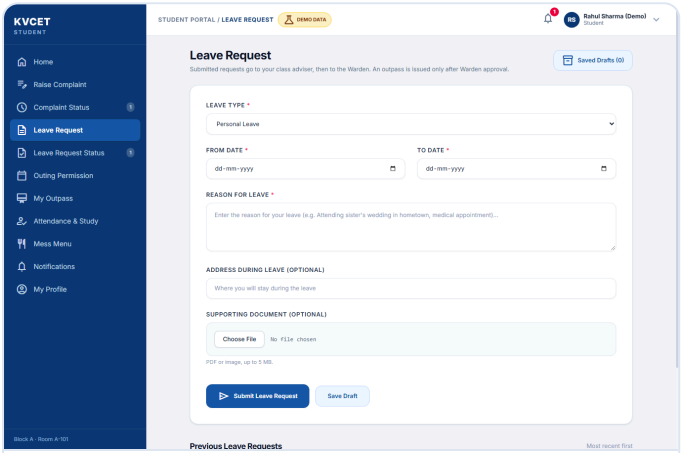
SYSTEM
4. Outpass issued
QR code sent to the student

SECURITY
5. Gate scan
Exit, then return

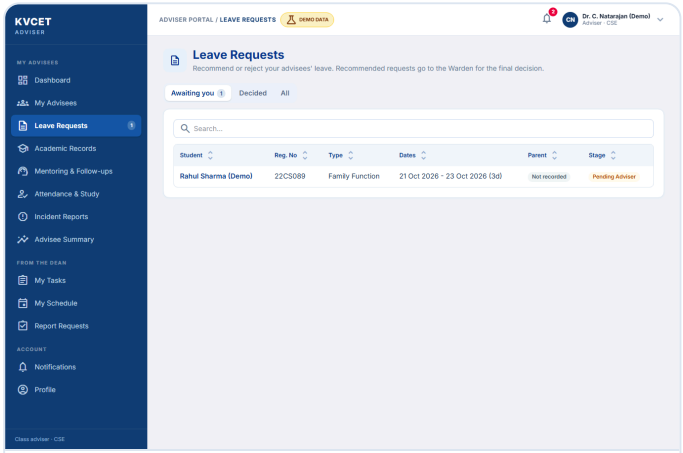
Leave status Pending Adviser → Pending Warden → Approved or Rejected / Cancelled

Outpass Active → Out → Completed or Cancelled

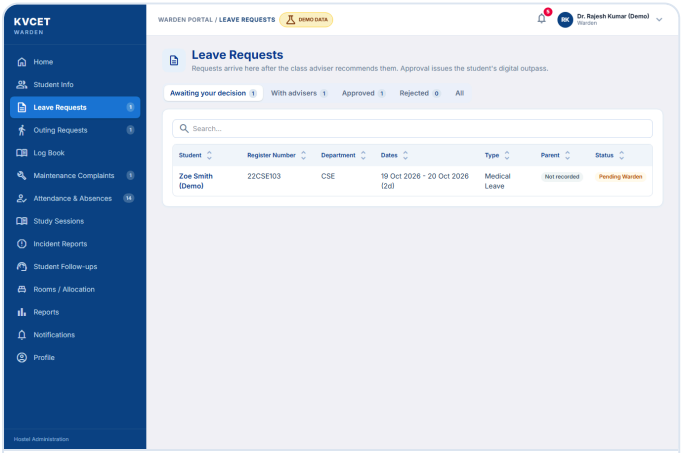
- Overlapping or duplicate leave requests are refused.
- The warden cannot approve before the adviser has recommended, and a decision cannot be changed once made.
- If a student exits without a valid pass, the gate entry is **flagged** and the warden is alerted.
- While the student is away, the attendance roll shows them as **On Leave**, so they are not counted as absent.



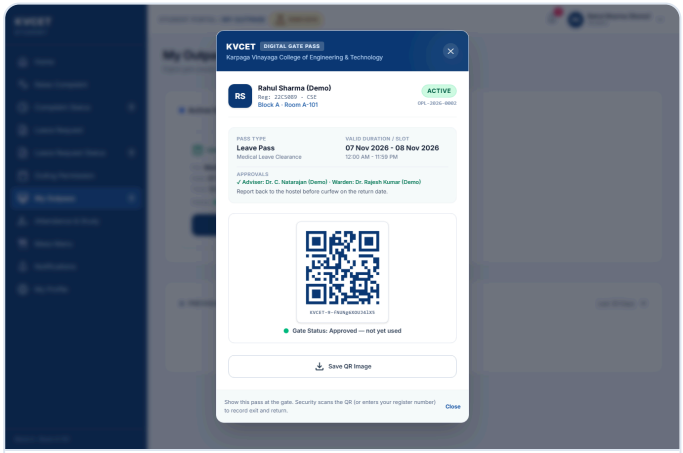
Student: leave request form



Adviser: recommend or reject advisee leave



Warden: requests awaiting the final decision

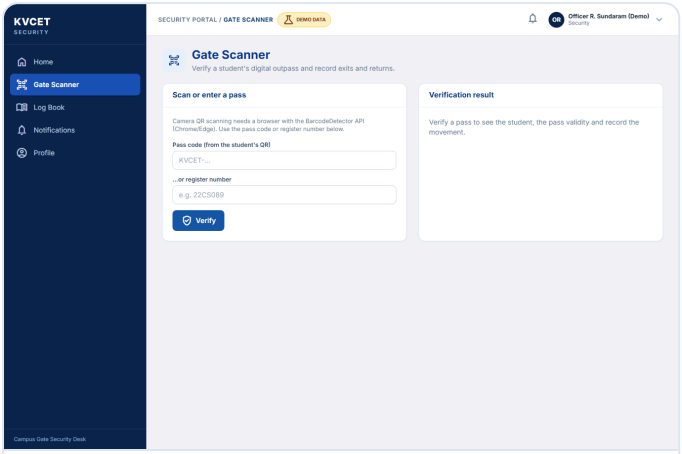


Student: approved outpass with QR code

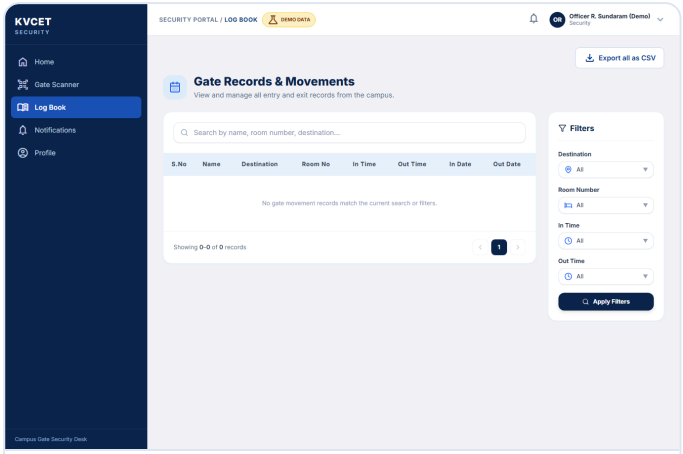
Gate: Security portal

Security scans the outpass QR code with the device camera, or types the pass code or register number. The system checks that the pass belongs to that student, is approved, and is within its valid dates. It then records the exit or return. Every movement appears in the gate log book, which the warden also sees.

Gate entry Approved Flagged Duplicate



Security: QR scanner with manual entry fallback



Security: gate log book

4 · OUTING

Outing permission

Outings are short trips on the same day. They need only the warden's approval.

STUDENT
1. Request
Date, time slot, place, purpose. Can be saved as a draft.

›

WARDEN
2. Approve / reject

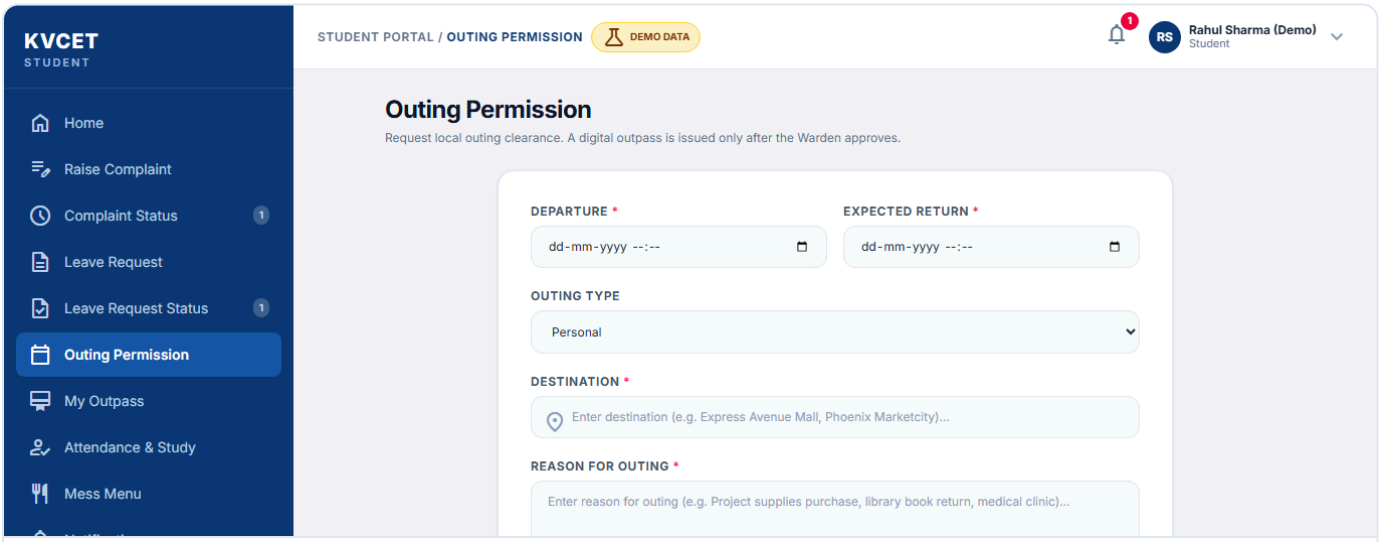
›

SYSTEM
3. Outing pass
QR code for that time slot

›

SECURITY
4. Exit & return
A late return is visible to the warden

Outing status Draft → Pending → Approved or Rejected / Cancelled



Student: outing permission request

5 · ATTENDANCE

Hostel attendance and unauthorized absence

The warden takes the hostel roll. Students on approved leave or outing are filled in automatically. Any other absence becomes a case that has to be verified and closed.

WARDEN
1. Take roll
Present / Absent. On Leave and On Outing come from approved passes.

›

SYSTEM
2. Close roll
Absent with no pass becomes *Suspected Unauthorized*

›

WARDEN
3. Verify
Authorized absence, or Verified Unauthorized

›

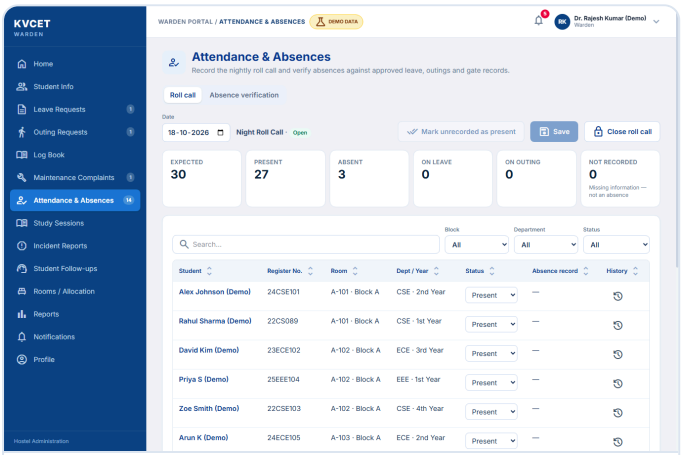
DEAN
4. Monitor
Requests a follow-up

›

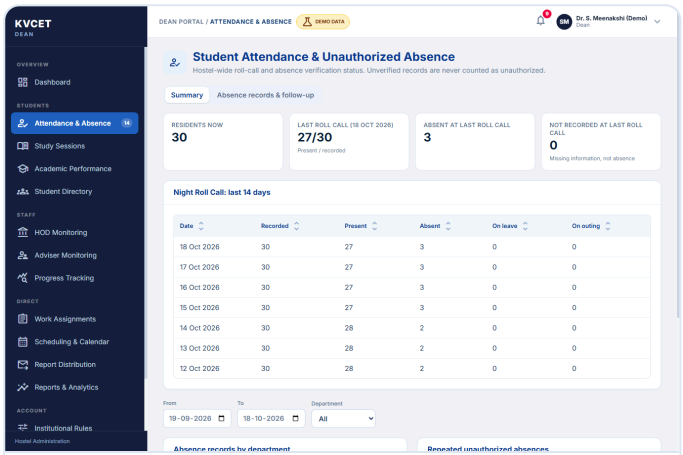
ADVISER **HOD**
5. Resolve
Records the outcome

Absence case Pending Verification → Authorized Absence or Suspected Unauthorized → Verified Unauthorized → Resolved

- Students see their own attendance in the student portal.
- Students with repeated absences are listed for the Dean, Adviser and HOD, using the thresholds set under **Institutional Rules**.



Warden: attendance roll and absence cases



Dean: attendance and unauthorized absence monitoring

6 · STUDY SESSIONS

Study sessions

WARDEN
1. Schedule
Date, time, venue

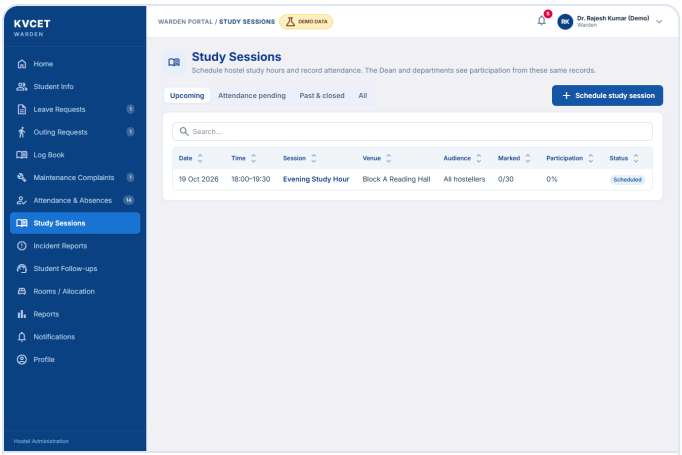
›

WARDEN
2. Mark attendance
Present / Late / Absent / Excused

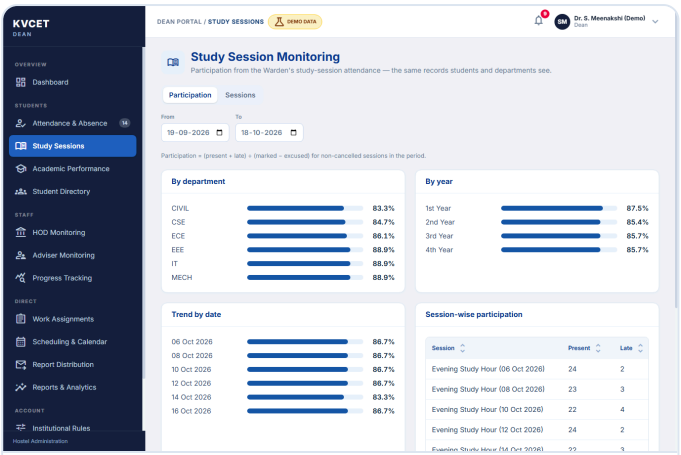
›

DEAN **HOD** **ADVISER**
3. Participation
Same records, plus repeated misses

Session Scheduled → Completed or Cancelled



Warden: study sessions and attendance marking



Dean: study-session participation; click a student to request a follow-up

7 · COMPLAINTS

Complaint → maintenance work order

A student's complaint is the same record the warden triages and the AO turns into a work order. When the work is completed, the student sees the resolution.

STUDENT
1. Raise
Category, room, description

›

WARDEN
2. Triage
Resolve directly, create a work order, or escalate

›

AO
3. Work order
Assign a technician and track it

›

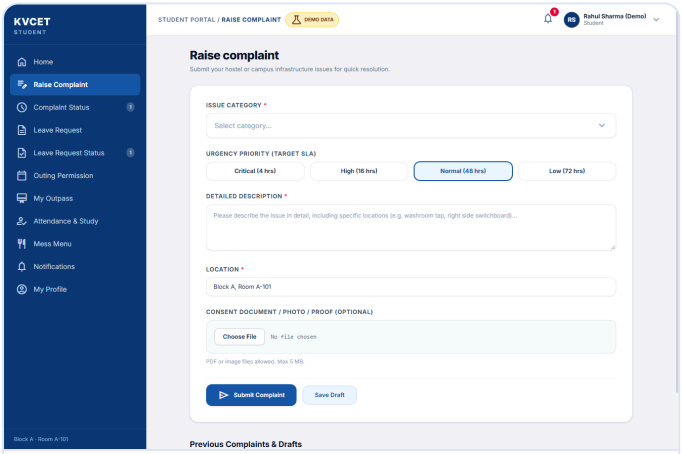
AO
4. Complete
Verify the work

›

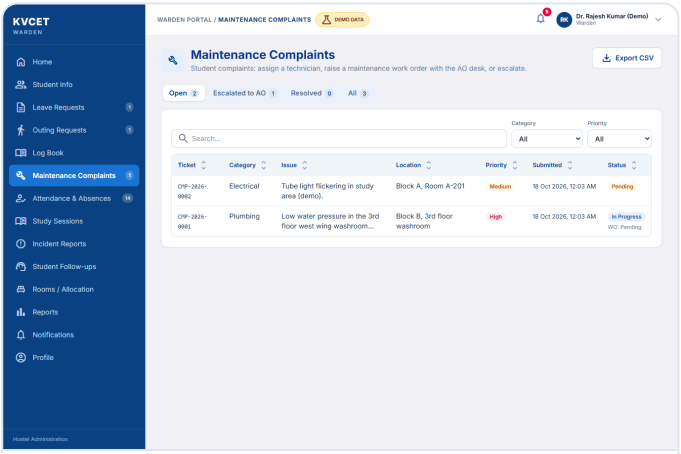
STUDENT
5. Resolved
Visible in Complaint Status

Complaint Pending → In Progress → Resolved or Escalated

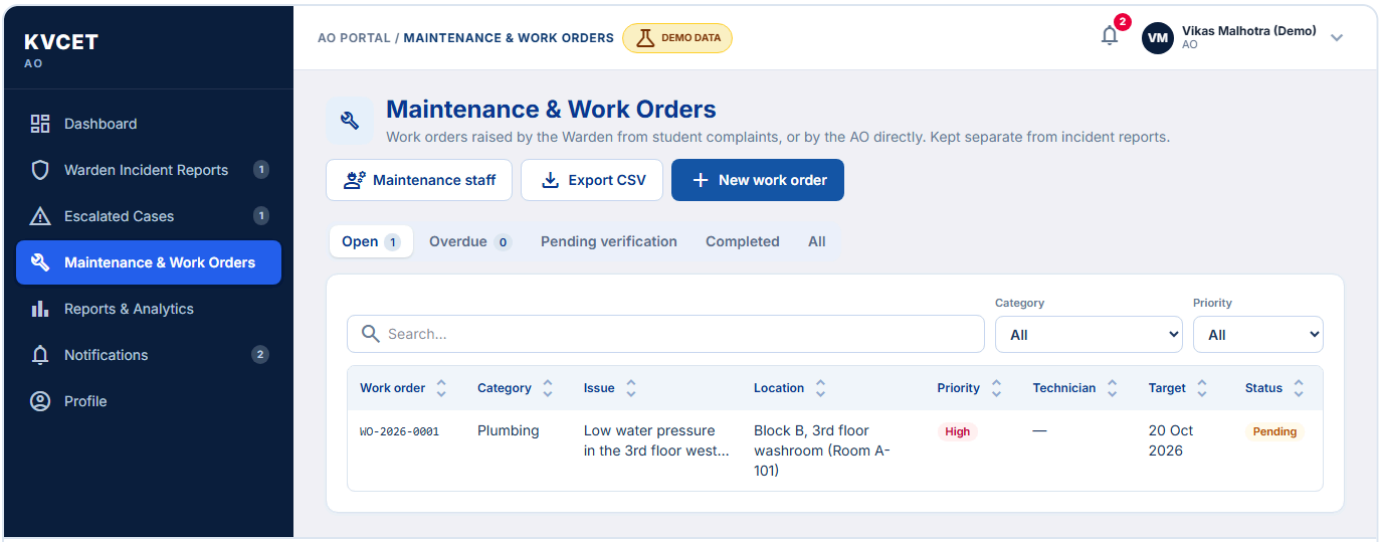
Work order Pending → Assigned → In Progress → Pending Verification → Completed



Student: raise a complaint



Warden: maintenance complaints



AO: maintenance and work orders

8 · INCIDENTS

Incident reports and AO escalation

WARDEN

1. File incident

Category, severity, students involved

AO

2. Review

Investigate or escalate

AO

3. Escalated case

Review, forward to the Principal, resolve

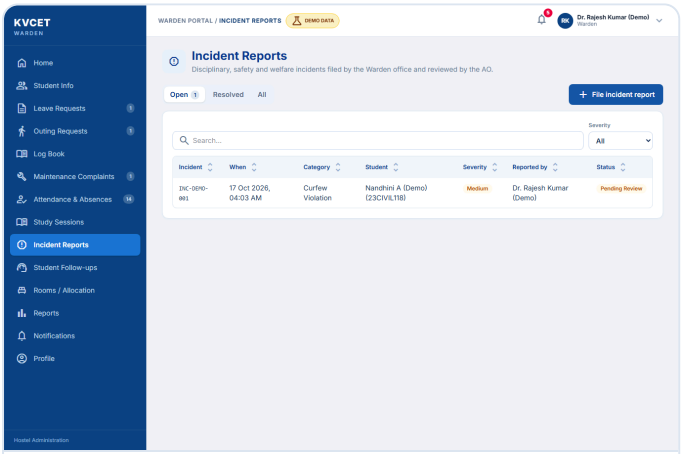
HOD **ADVISER**

4. Informed

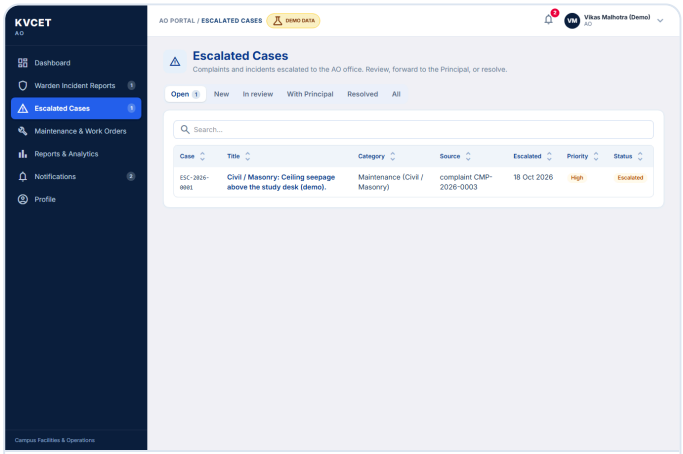
See cases for their own students only

Incident Pending Review → Under Investigation → Escalated / Resolved

Escalation Escalated → In Review → Forwarded to Principal → Resolved



Warden: incident reports



AO: escalated cases

9 · ACADEMICS

Academic performance and student follow-ups

Marks come only from imported records; nothing is estimated. Rules set by the Dean flag students who need support, and each flagged student is handled through a tracked follow-up.

ADVISER **HOD**

1. Import marks

CSV, checked row by row, all-or-nothing

›

SYSTEM

2. Apply rules

Below threshold, declining, or arrears → *Needs support*

›

DEAN

3. Request follow-up

From the student profile

›

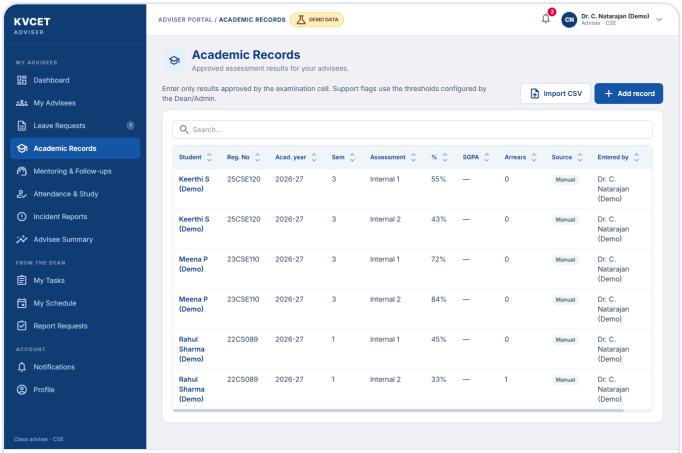
ADVISER **HOD** **WARDEN**

4. Act

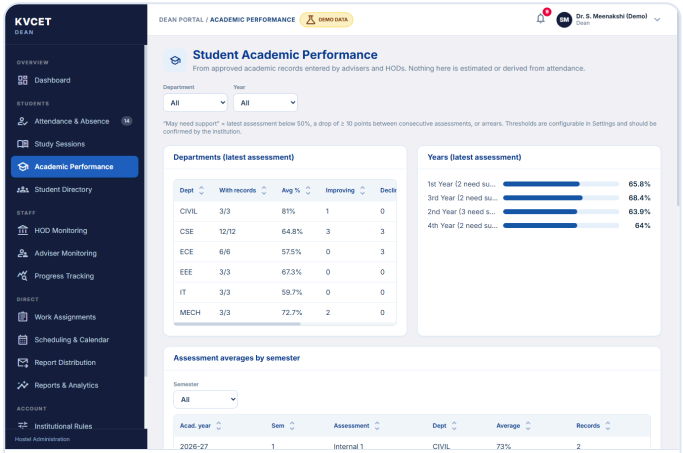
Mentoring session, outcome, close

Follow-up Open → In Progress → Completed or Cancelled

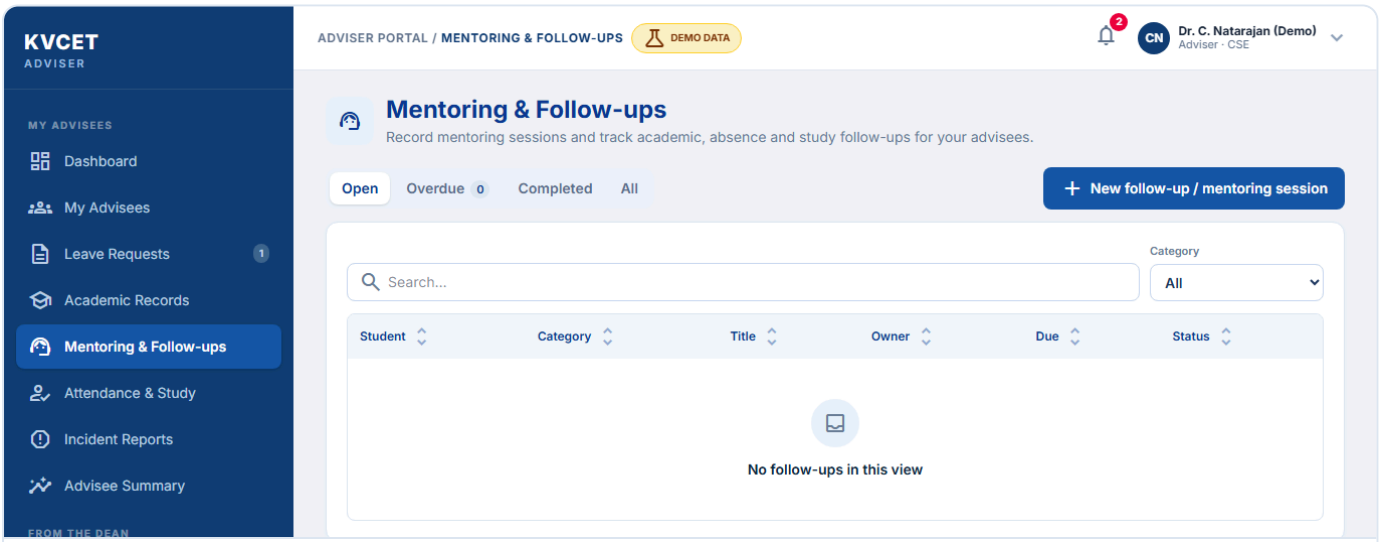
- An adviser can import marks only for their own advisees, and an HOD only for their own department.
- The Dean can assign a follow-up only to the student's adviser, their department's HOD, or a warden.
- HOD and Adviser monitoring shows workload and pending items, never a personal score.



Adviser: academic records and import



Dean: academic performance across departments



Adviser: mentoring and follow-ups

10 · DEAN WORK

Dean work assignments

The Dean sends a task to one or more HODs or Advisers. Each recipient has their own status, so one late recipient does not hide the progress of the others.

DEAN
1. Create
Title, priority, deadline, whether evidence is required

›

DEAN
2. Confirm recipients
Preview the list, then confirm the count

›

HOD **ADVISER**
3. Acknowledge & work
Progress updates

›

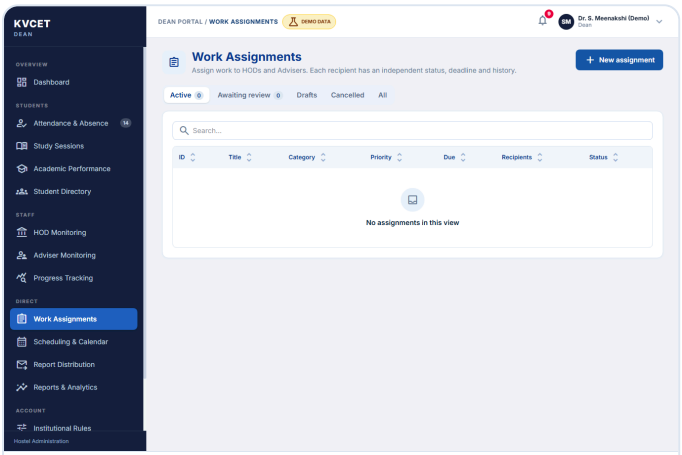
HOD **ADVISER**
4. Submit
With an attachment if evidence is required

›

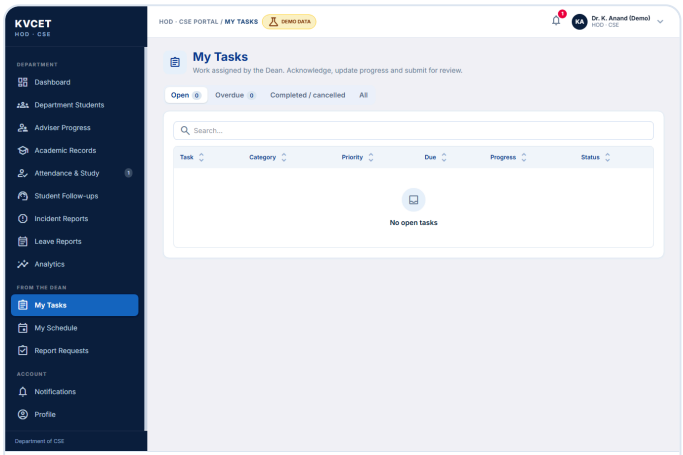
DEAN
5. Review
Approve, or request a revision

Per recipient Assigned → Acknowledged → In Progress → Submitted → Revision Requested → Submitted → Completed

- Reminders are sent automatically before the deadline and when it is overdue, once each.
- Each resubmission increases the version number, and the full history is kept.



Dean: assignments with per-recipient progress



HOD: task inbox

11 · SCHEDULING & REPORTS

Dean scheduling

DEAN
1. Create event
Meeting, review, inspection

›

SYSTEM
2. Conflict check
An overlap needs an override reason

›

HOD **ADVISER**
3. Acknowledge
Appears in their My Schedule page

›

DEAN
4. Reschedule / cancel
Participants are notified

Event Scheduled → Rescheduled → Completed or Cancelled

Report distribution

DEAN
1. Draft request
Report type, deadline, instructions

›

DEAN
2. Distribute
After the recipient preview

›

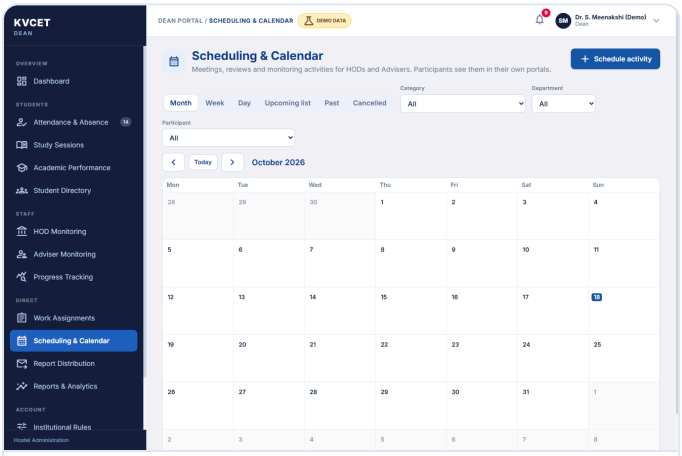
HOD **ADVISER**
3. Acknowledge & submit

›

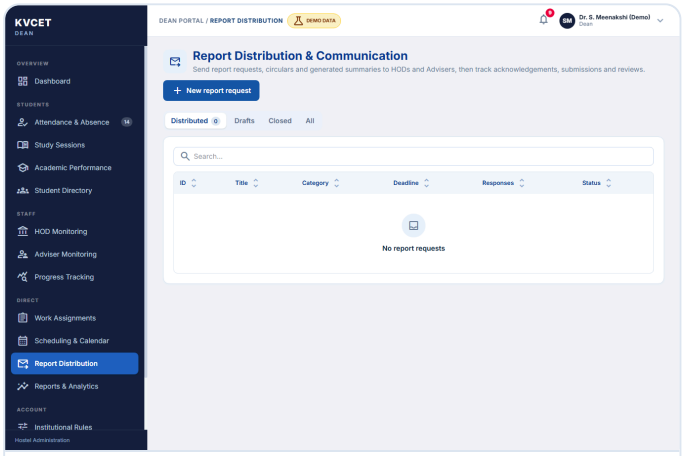
DEAN
4. Review
Accept or request a revision, then close

Per recipient Distributed → Acknowledged → Submitted → Under Review → Revision Requested → Accepted

Request Draft → Distributed → Closed



Dean: schedule and calendar



Dean: report distribution

12 · MESS

Mess menu and feedback

MESS
1. Draft menu
Weekly, by meal

›

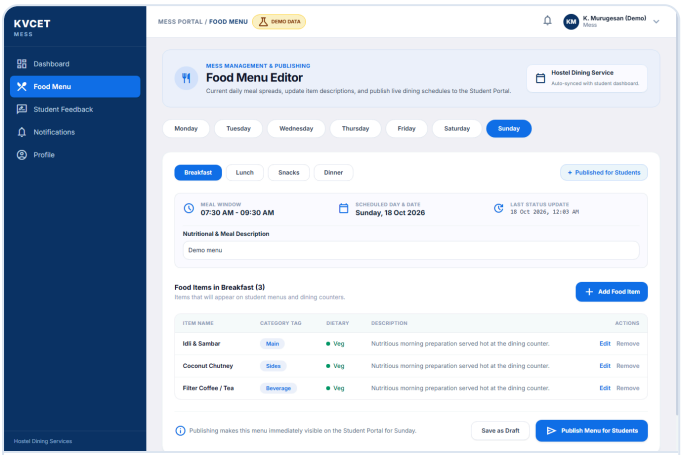
MESS
2. Publish
Students see only the published menu

›

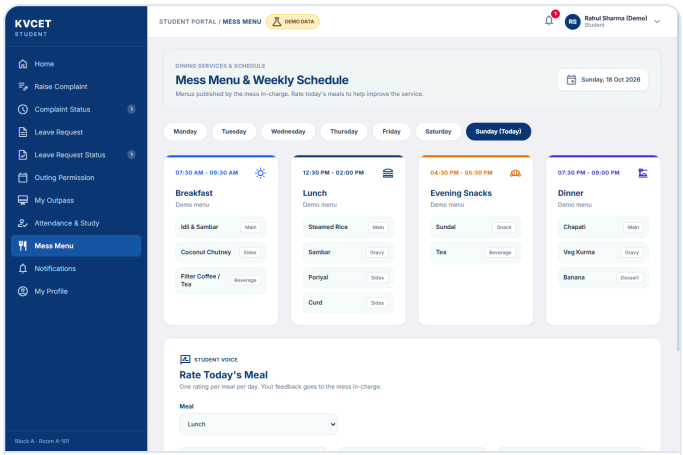
STUDENT
3. View & rate
One rating per meal per day

›

MESS
4. Review feedback



Mess: food menu (draft and publish)

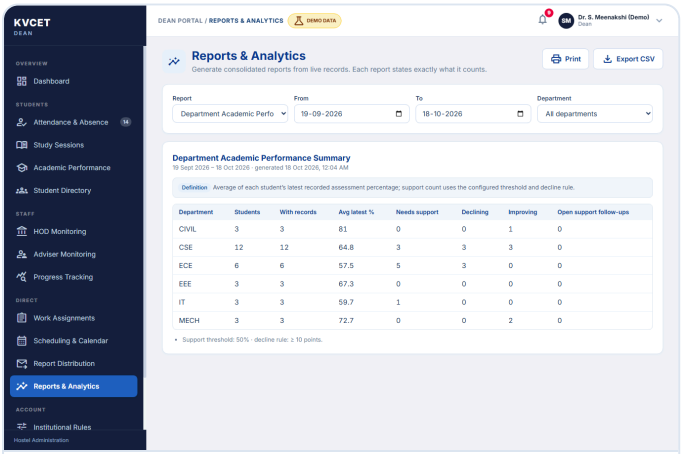


Student: published mess menu

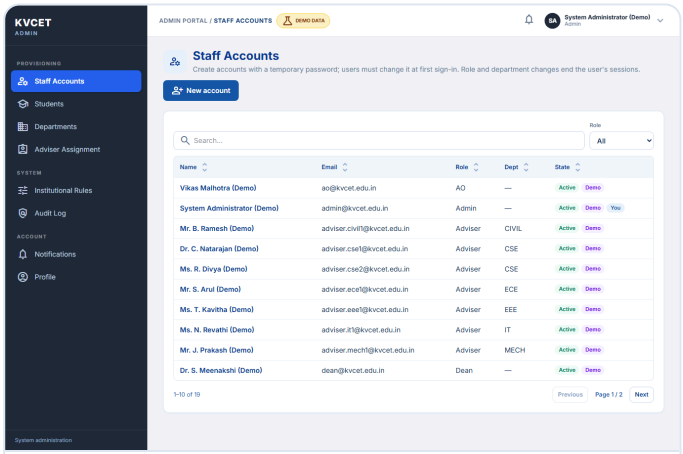
13 · NOTIFICATIONS, REPORTS, ADMINISTRATION

Notifications, reports and administration

- **Notifications:** the bell in every portal shows in-app notifications. Each one links to its record, and duplicates are never sent. Email and SMS are not connected.
- **Reports & analytics:** the Warden, HOD, Adviser, AO and Dean can generate reports within their own scope. Reports can be downloaded as CSV or printed.
- **Institutional Rules:** the Dean or Admin sets curfew, attendance thresholds, academic-support thresholds and study-session rules.
- **Administration:** the Admin creates accounts, students, departments and rooms, assigns advisers, resets passwords and reviews the audit log.



Dean: reports and analytics with CSV and print



Admin: staff accounts

14 · REFERENCE

Portal menu reference

Portal	Sidebar menu
STUDENT	Home · Raise Complaint · Complaint Status · Leave Request · Leave Request Status · Outing Permission · My Outpass · Attendance & Study · Mess Menu · Notifications · My Profile
ADVISER	Dashboard · My Advisees · Leave Requests · Academic Records · Mentoring & Follow-ups · Attendance & Study · Incident Reports · Advisee Summary · My Tasks · My Schedule · Report Requests · Notifications · Profile
HOD	Dashboard · Department Students · Adviser Progress · Academic Records · Attendance & Study · Student Follow-ups · Incident Reports · Leave Reports · Analytics · My Tasks · My Schedule · Report Requests · Notifications · Profile
WARDEN	Home · Student Info · Leave Requests · Outing Requests · Log Book · Maintenance Complaints · Attendance & Absences · Study Sessions · Incident Reports · Student Follow-ups · Rooms / Allocation · Reports · Notifications · Profile
SECURITY	Home · Gate Scanner · Log Book · Notifications · Profile
AO	Dashboard · Warden Incident Reports · Escalated Cases · Maintenance & Work Orders · Reports & Analytics · Notifications · Profile
MESS	Dashboard · Food Menu · Student Feedback · Notifications · Profile
DEAN	Dashboard · Attendance & Absence · Study Sessions · Academic Performance · Student Directory · HOD Monitoring · Adviser Monitoring · Progress Tracking · Work Assignments · Scheduling & Calendar · Report Distribution · Reports & Analytics · Institutional Rules · Notifications · Profile
ADMIN	Staff Accounts · Students · Departments · Adviser Assignment · Institutional Rules · Audit Log · Notifications · Profile

End-to-end example: one leave, every portal

#	Who	What happens	Where it shows up
1	STUDENT	Submits leave `LR-2026-0004` for 2 days	Adviser's Leave Requests; notification to the adviser
2	ADVISER	Recommends it	Warden's "Awaiting your decision"; the student sees *Pending Warden*
3	WARDEN	Approves it	The student gets a QR outpass; Security can now validate it
4	SECURITY	Scans the QR at exit, and again at return	Gate log book for Security and Warden; outpass becomes Out, then Completed
5	WARDEN	Takes the night roll	The student shows as *On Leave*, not absent
6	DEAN	Opens Attendance monitoring	Counts the leave as an authorized absence

Screenshots in this guide come from the application running with **demo data** (names marked "(Demo)"). They are examples, not real student records.